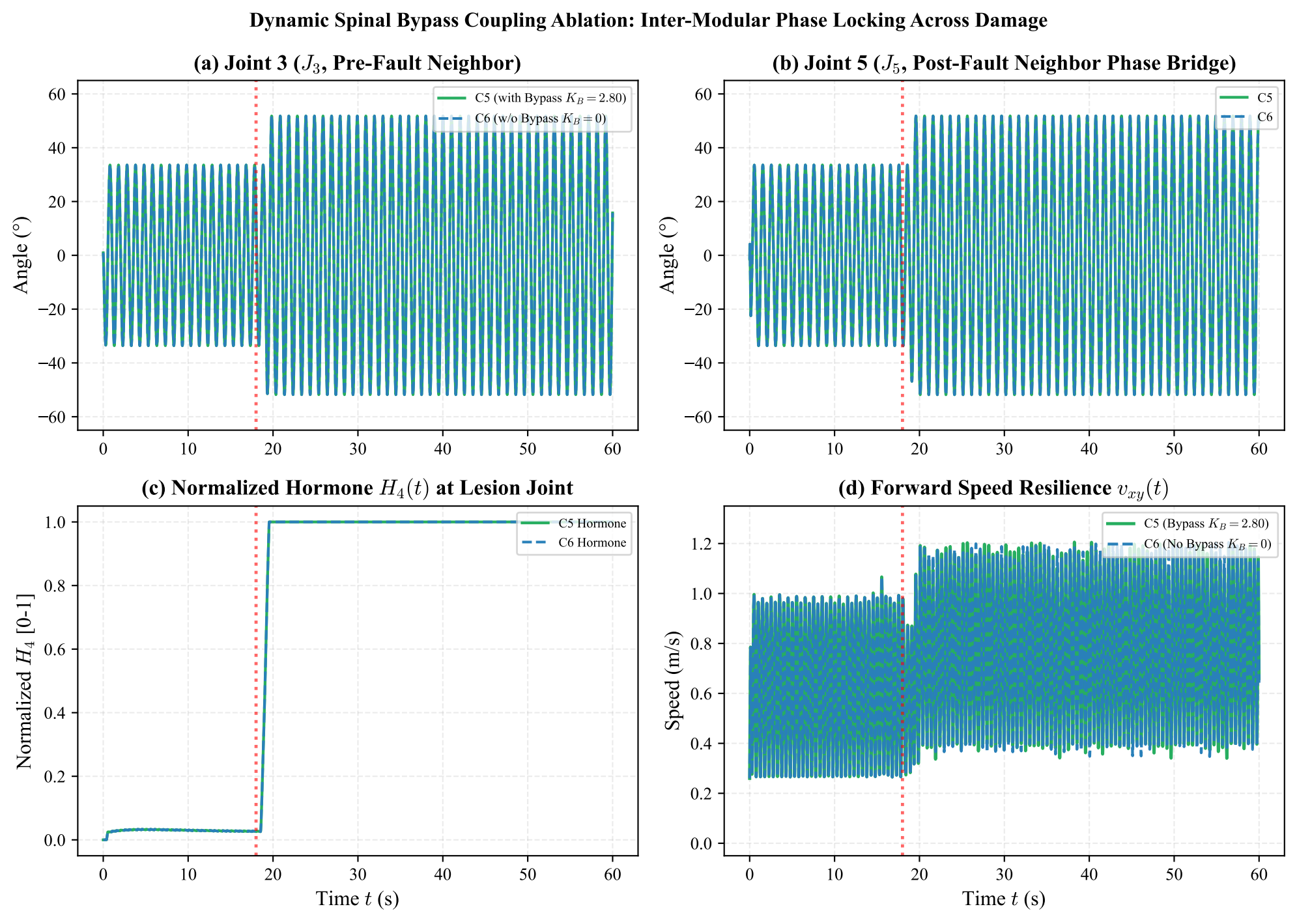

Dynamic Spinal Bypass Coupling Ablation: Inter-Modular Phase Locking Across Damage
(a) Joint 3 (J_3, Pre-Fault Neighbor)
C5 (with Bypass $K_B = 2.80$)
C6 (w/o Bypass $K_B = 0$)
Angle (°)
(b) Joint 5 (J_5, Post-Fault Neighbor Phase Bridge)
C5
C6
Angle (°)
(c) Normalized Hormone $H_4(t)$ at Lesion Joint
C5 Hormone
C6 Hormone
Normalized H_4 [0-1]
Time t (s)
(d) Forward Speed Resilience $v_{xy}(t)$
C5 (Bypass $K_B = 2.80$)
C6 (No Bypass $K_B = 0$)
Speed (m/s)
Time t (s)